

PAT 6 - Ratios & Percentage

Name: _____

Date: _____

1. A recipe for lemonade calls for 1 cup of sugar and 5 cups of water. How much sugar is used per cup of water?

A. $\frac{1}{6}$ cup B. $\frac{1}{5}$ cup C. $\frac{1}{4}$ cup D. $\frac{5}{1}$ cup

2. A high-speed elevator can rise 480 feet in 30 seconds. Which expression represents the rate, in feet per minute, of the elevator?

A. 480×30 B. $480 \div 30$ C. 480×2 D. $480 \div 2$

3. The area of an airplane's wings is related to the airplane's lifting force, which holds the airplane in the air. The table below lists several wing areas and the corresponding lifting forces.

AIRPLANE LIFTING FORCE

Area of Wings (square feet)	Lifting Force (pounds)
125	1,875
150	2,250
175	2,625
250	3,750
x	5,625
420	y

The ratio of lifting force to area is equivalent for all pairs in the table. What are the values of x and y ?

- A. $x = 375$ square feet and $y = 7,500$ pounds B. $x = 335$ square feet and $y = 7,500$ pounds
C. $x = 375$ square feet and $y = 6,300$ pounds D. $x = 335$ square feet and $y = 6,300$ pounds

4. A group of students organized a car wash to raise money for a local charity. The students charged \$5.00 for each car they washed. In 3 hours, they washed 12 cars. At that rate, how much money could they earn from washing cars for eight hours?

A. \$40.00 B. \$60.00 C. \$85.00 D. \$160.00

5. Simon used 3 pears and 9 apples to make a fruit salad. What was the ratio of the number of pears to the number of apples in the fruit salad?

A. 1 : 3 B. 1 : 4 C. 1 : 6 D. 1 : 9

6. A basketball player attempts 15 baskets in a game. He makes 9 of the attempted baskets. Which ratio describes the number of baskets the player made to the number of baskets the player attempted?
- A. $\frac{3}{5}$ B. $\frac{5}{3}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$
7. Wyatt hiked 6 miles in 2 hours. At this same rate, what is the total number of miles Wyatt could hike in 9 hours?
- A. 3 B. 7 C. 21 D. 27
8. A bakery sells 5 apple muffins for every 2 bran muffins sold. Which table shows this ratio?
- A.

Apple	Bran
5	2
10	12
20	22
- B.

Apple	Bran
10	4
15	6
35	14
- C.

Apple	Bran
5	2
18	8
20	10
- D.

Apple	Bran
20	4
30	6
40	8
9. A punch recipe requires 2 cups of cranberry juice to make 3 gallons of punch. Using the same recipe, what is the amount of cranberry juice needed for 1 gallon of punch?
- A. 3 cups B. $1\frac{1}{2}$ cups C. 1 cup D. $\frac{2}{3}$ cup
10. Arnold's entire workout consisted of 10 minutes of warm-up exercises, 25 minutes of lifting weights, and 15 minutes on the treadmill. What was the ratio of the number of minutes he lifted weights to the total number of minutes of his entire workout?
- A. 1:1 B. 1:2 C. 3:10 D. 5:8
11. The water level in an ocean bay changes at an average rate of 3 meters per hour. At this rate, how many hours would it take for the water level to change 12 meters?
- A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. 4 D. 36
12. In Ms. Perron's class, 75% of the students are boys. There are 18 boys in the class. What is the total number of students in Ms. Perron's class?
- A. 6 B. 14 C. 24 D. 57
13. All the students in the sixth grade either purchased their lunch or brought their lunch from home on Monday.
- 24% of the students purchased their lunch.
 - 190 students brought their lunch from home.

How many students are in the sixth grade?

- A. 76 B. 166 C. 214 D. 250

14. Residents of a small city voted on whether to allow a developer to build a shopping center. The number of votes in favor of the shopping center was 4,400. The number of votes against the shopping center was 17,600. What percent of the voters were *in favor* of building the shopping center?
- A. 20% B. 25% C. 40% D. 44%
15. There are 230 calories in 4 ounces of a type of ice cream. How many calories are in 6 ounces of that ice cream?
- A. 232 B. 236 C. 345 D. 460
16. The table below shows different possibilities for the number of games a team would need to win to maintain a certain percentage of wins.

POSSIBLE BASEBALL GAMES WON

Number of Games Won	Number of Games Played
6	10
24	40
36	60
42	70

Which ratio of the number of games won to the number of games played could also be included in this table?

- A. 18:20 B. 30:20 C. 18:30 D. 50:30
17. Last year the girls' basketball team had 8 fifth-grade students and 7 sixth-grade students. What was the ratio of sixth-grade students to fifth-grade students on the team?
- A. 8:15 B. 8:7 C. 7:8 D. 15:8
18. Sam paid \$8.28 for 18 stamps. At this rate, how much would it cost Sam to buy 12 stamps?
- A. \$2.19 B. \$2.28 C. \$3.72 D. \$5.52
19. Pat bounces a basketball 25 times in 30 seconds. At that rate, approximately how many times will Pat bounce the ball in 150 seconds?
- A. 120 B. 125 C. 144 D. 145
20. Kate has a coin collection. She keeps 7 of the coins in a box, which is only 5% of her entire collection. What is the total number of coins in Kate's coin collection?
- A. 12 B. 14 C. 120 D. 140

21. The table below lists four masses and their corresponding approximate weights on Earth.

MASSES AND CORRESPONDING WEIGHTS

Mass (kilograms)	Weight (Newtons)
20	196
50	490
x	1078
130	1274
140	1372

The ratio of weight to mass is constant. Which statement describes the ratio of weight to mass and the value of x in the table?

- A. The ratio is $\frac{98}{10}$; $x = 90$
- B. The ratio is $\frac{98}{10}$; $x = 110$
- C. The ratio is $\frac{10}{98}$; $x = 90$
- D. The ratio is $\frac{10}{98}$; $x = 110$
22. Atlantic Auditorium has 850 seats. Tickets were sold for 816 of the seats. For what percent of the seats were tickets sold?
23. A grocery store sells a bag of 5 lemons for \$2.00. What is the unit cost of each lemon in the bag?
- A. \$2.50
- B. \$0.60
- C. \$0.40
- D. \$0.10
24. At a concert, 20% of the audience members were teenagers. If the number of teenagers at the concert was 360, what was the total number of audience members?
- A. 432
- B. 450
- C. 1,800
- D. 7,200
25. Jaden made a pot of chili with 48 ounces of ground beef and 2 tablespoons of chili powder. He made another pot of chili with the same amount of ground beef, but he used 3 times as much chili powder. How many pounds of ground beef per tablespoon of chili powder did he use in the second pot of chili?